Statistical Thinking for Data Science

Week 7: Probability as a language for uncertainty

Saying how often, before you have to say how sure

Dr Abdulzeid Yen Anafo

Department of Computing and Data Analytics
University of Mines and Technology, Tarkwa

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Lecture notes for students of Statistical Thinking for Data Science.

Week 7: Probability as a language for uncertainty

Learning Objectives

By the end of this week you should be able to:

- write down a sample space and compute probabilities of events from the axioms;
- explain what a probability means, under both the long run and the degree of belief reading, and say which one a given sentence uses;
- define a random variable and distinguish its distribution from the data;
- compute the mean and standard deviation of a discrete distribution;
- choose between the binomial, Poisson and normal distributions from the mechanism that generated the data, not from the shape of a histogram;
- use the normal distribution to find probabilities and percentiles;
- estimate a probability by simulation when the algebra is hard, and say how accurate the estimate is.

7.1 Why a whole language for uncertainty

Everything so far has described data we already have. From here on we reason about data we do not have: the next sample, the rest of the population, next month's production. That step always carries uncertainty, and probability is the language engineers and scientists use to carry it precisely rather than with words like "probably" and "unlikely", which different readers price differently.

Two readings of one number

"The probability of a misfire is 0.03" can mean two things. **Long run frequency**: in many repetitions, about 3% of detonators misfire. **Degree of belief**: given what I know, I would accept a bet at those odds. The arithmetic is identical under both readings, which is why the same rules serve a quality engineer and a judge. The readings differ when the event happens only once, and we return to that in Week 12.

7.2 Sample spaces, events and the three axioms

An **experiment** is any process with an uncertain outcome. The **sample space** S is the set of all outcomes, and an **event** is a subset of S. For two tosses of a coin, $S = \{HH, HT, TH, TT\}$, and "at least one head" is $\{HH, HT, TH\}$.

A probability is a function P on events obeying three rules:

1. $P(A) \geq 0$ for every event A;
2. $P(S) = 1$;
3. if A and B cannot both happen, $P(A \cup B) = P(A) + P(B)$.

Everything else follows. $P(A^c) = 1 - P(A)$. $P(\emptyset) = 0$. For any two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

because adding the two probabilities counts the overlap twice.

Example 1: two dice

Two fair dice are thrown. Find the probability that the total is at least 10, and the probability

that at least one die shows a six.

Solution

There are 36 equally likely outcomes. Totals of 10 or more come from $(4,6),(5,5),(6,4),(5,6),(6,5),(6,6)$: six outcomes, so $6/36 = 1/6 = 0.167$.
For at least one six, it is quicker to use the complement. No six at all has $5 \times 5 = 25$ outcomes, so $P(\text{at least one six}) = 1 - 25/36 = 11/36 = 0.306$.

The habit

"At least one" almost always means "one minus the probability of none". You will use this constantly.

Example 2: shared birthdays, and why intuition fails

In a tutorial group of 23, what is the probability that two people share a birthday?

Solution

Compute the complement: all 23 birthdays different. The first person is free, the second must avoid one day, the third two, and so on:

$$P(\text{all different}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{343}{365} = 0.4927,$$

so the probability of a shared birthday is $1 - 0.4927 = 0.5073$, slightly better than even. With 50 people it is 0.9704.

Why it surprises

People think of the 22 comparisons involving themselves. The event concerns all $\binom{23}{2} = 253$ pairs. Ask what is being compared with what: that is the same question as last week's *compared to what*, in a different costume.

7.3 Random variables

A **random variable** is a rule that attaches a number to each outcome. The number of faulty cubes in a batch of 12, the time until a pump fails, the grade of the next core sample: each is a function from the sample space to the real line.

The **distribution** of a random variable says how probability is spread over its possible values. Keep the distinction sharp:

	Data	**Distribution**
What it is	numbers you have measured	a model for how such numbers arise
Centre	sample mean $\bar{x}$	expected value $\mu = \mathrm{E}(X)$
Spread	sample sd s	$\sigma = \sqrt{\mathrm{Var}(X)}$
Picture	histogram	probability mass or density function
Changes when	you collect more data	you change the model

For a discrete random variable with mass function $p(x)$,

$$\mathrm{E}(X) = \sum_x x\,p(x), \qquad \mathrm{Var}(X) = \sum_x (x-\mu)^2 p(x) = \mathrm{E}(X^2) - \mu^2.$$

Example 3: stoppages in a shift

The number of unplanned stoppages X in a shift has $p(0) = 0.50$, $p(1) = 0.30$, $p(2) = 0.15$, $p(3) = 0.05$.

Solution

$\mathrm{E}(X) = 0(0.50) + 1(0.30) + 2(0.15) + 3(0.05) = 0.75$ stoppages per shift.
$\mathrm{E}(X^2) = 0 + 0.30 + 4(0.15) + 9(0.05) = 1.35$, so $\mathrm{Var}(X) = 1.35 - 0.75^2 = 0.7875$ and $\sigma = 0.887$.

Reading it

The mean, 0.75, is not a possible value: no shift ever has three quarters of a stoppage. It is the long run average, and it is exactly what you need to plan maintenance hours over a month, for the same reason the mean repair time was the planner's number in Week 6.

7.4 Three distributions worth knowing by mechanism

Do not choose a distribution by looking at a histogram and picking the closest shape. Choose it by asking what generated the numbers. Each of the three below is the answer to a specific mechanism, and if the mechanism does not match, neither will the model.

7.4.1 Binomial: counting successes in a fixed number of independent trials

Conditions: a fixed number n of trials; each trial succeeds or fails; the same probability p every time; trials independent. Then

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad \mathrm{E}(X) = np, \qquad \sigma = \sqrt{np(1-p)}.$$

Example 4: cubes from the Week 5 supplier

Twelve cubes are tested, and each fails independently with probability 0.09.

Solution

$P(\text{none fail}) = 0.91^{12} = 0.3225$. $P(\text{exactly one}) = 12(0.09)(0.91)^{11} = 0.3827$. $P(\text{two or more}) = 1 - 0.3225 - 0.3827 = 0.2948$. The mean is $12(0.09) = 1.08$ failures and $\sigma = \sqrt{12(0.09)(0.91)} = 0.99$.

Where it breaks

If the twelve cubes come from one truckload mixed on one bad morning, the trials are not independent and the binomial understates the chance of several failures together. The arithmetic will still produce a number. That is the danger.

7.4.2 Poisson: counting events in a window of time or space

Conditions: events happen at a constant average rate; they are independent; two cannot occur at the same instant. Then with mean rate λ per window,

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad \mathrm{E}(X) = \lambda, \qquad \sigma = \sqrt{\lambda}.$$

Trucks arriving at a weighbridge, flaws per kilometre of weld, calls to a control room: all Poisson if the rate really is steady. The variance equals the mean, which gives a quick check: if your counts scatter far more than that, the rate is not constant and you need a different model.

Example 5: trucks at the weighbridge

Trucks arrive at 2.5 per ten minutes on average.

Solution

$P(\text{none in ten minutes}) = e^{-2.5} = 0.0821$. $P(\text{five or more}) = 1 - \sum_{k=0}^{4} e^{-2.5} 2.5^k / k! = 0.1088$. So about one ten minute window in twelve is empty, and about one in nine is busy. Staffing for the average, 2.5, leaves you overwhelmed in one window in nine.

7.4.3 Normal: sums and averages of many small effects

Conditions: the quantity is produced by adding up many small independent contributions. Measurement error, total weight of a loaded truck, the mean of a large sample: all approximately normal, and we will see in Week 9 why the last one matters most.

The normal density is symmetric and bell shaped, described entirely by μ and σ. The useful facts:

- $P(|Z| < 1) = 0.6827$, $P(|Z| < 2) = 0.9545$, $P(|Z| < 3) = 0.9973$;
- the central 95% sits within 1.96 standard deviations of the mean, and the central 90% within 1.645;
- standardising, $Z = (X - \mu)/\sigma$, turns any normal question into a standard normal one.

Example 6: a cohort of marks

Marks are modelled as normal with $\mu = 62$ and $\sigma = 8$.

Solution

$P(X > 75)$: $z = (75 - 62)/8 = 1.625$, giving $P = 0.0521$, about 5% of the class.
$P(50 < X < 70)$: $z = -1.5$ and $z = 1.0$, giving $0.8413 - 0.0668 = 0.7745$.
The 90th percentile: $z = 1.2816$, so $x = 62 + 1.2816(8) = 72.3$ marks.

A warning

Marks are bounded at 0 and 100 and are often skewed, so the normal model is an approximation that fails in the tails. It is fine for "roughly how many above 75" and wrong for "the chance somebody scores above 100", which it puts at a small positive number rather than zero.

Choosing by mechanism

Fixed number of yes or no trials, constant chance, independent: **binomial**. Counting events in a window at a steady rate: **Poisson**. Many small effects adding up, or an average of a decent sized sample: **normal**. If none of the mechanisms fits, say so and simulate instead of forcing a formula.

7.5 Simulation: when the algebra is hard

Most real questions have no clean formula. A simulation answers them by doing the experiment many times in software and counting.

The recipe:

1. write down the mechanism precisely, including every assumption;
2. generate one artificial dataset from it;
3. compute the quantity you care about;
4. repeat several thousand times and look at the distribution of the results.

Example 7: how accurate is a simulated probability?

You estimate a probability by simulating 10 000 times and counting, and the answer comes out near 0.20.

Solution

A simulated proportion is itself an average, so it has a standard error of $\sqrt{p(1-p)/m}$ with m the number of runs: $\sqrt{0.2(0.8)/10\,000} = 0.004$. So the estimate is good to about ± 0.008 with 95% confidence, which is two decimal places and no more. To gain one more decimal place you need a hundred times as many runs, because the error falls with $\sqrt{m}$.

The rule

Always report the simulation error alongside a simulated answer. A probability quoted as 0.2037 from 10 000 runs is claiming precision it does not have.

Simulation is also the honest way to check whether a formula applies: simulate the mechanism you believe, and compare the simulated distribution with the formula's. If they disagree, the assumption you doubted is the one to examine.

7.6 Doing it on a computer

In R: `dbinom`, `pbinom`, `dpois`, `pnorm`, `qnorm` for the formulas, and `sample`, `rbinom`, `rnorm`, `replicate` for simulation. In Python: `scipy.stats.binom`, `poisson`, `norm`, each with `.pmf`, `.cdf` and `.ppf`, and `numpy.random.default_rng()` for simulation. Set a seed, so that your numbers can be reproduced by somebody checking your work.

7.7 Glossary

Sample space	the set of all possible outcomes of an experiment
Event	a subset of the sample space
Random variable	a rule attaching a number to each outcome
Distribution	how probability is spread over the values of a random variable
Expected value	the long run average value, $\sum x\,p(x)$
Parameter	a number describing the distribution, such as λ or μ
Standardising	converting to $Z = (X - \mu)/\sigma$
Monte Carlo	estimating a quantity by repeated random simulation

7.8 Summary

Week 7 in six lines

1. Probability is the language for data you do not have yet.
2. Three axioms generate every rule you will use; "at least one" means one minus none.
3. A distribution is a model, not a dataset: keep μ and $\bar{x}$ apart.
4. Choose binomial, Poisson or normal from the mechanism, never from the shape.
5. The normal has 68, 95 and 99.7 within one, two and three standard deviations.
6. If the algebra is hard, simulate, and report the simulation error.

Exercises

1. Two dice are thrown. Find $P(\text{total} = 7)$, $P(\text{total} \geq 10)$ and $P(\text{at least one even})$. For which of the three was the complement the easier route?

2. In a group of 50 students, find the probability that at least two share a birthday. Explain to a classmate why the answer is so much larger than the $50/365$ they guessed.

3. A machine makes parts with a 4% defect rate, independently. In a box of 20, find the probability of no defects, exactly one, and more than two. State the mean and standard deviation of the number of defects.

4. Flaws occur in a weld at 1.8 per metre. Find the probability of no flaw in a metre, and of three or more in two metres. State the assumption you made in going from one metre to two.

5. Marks are normal with mean 62 and standard deviation 8. Find the proportion above 75, the proportion between 50 and 70, and the mark that only the top 10% exceed. Then give one reason the normal model is wrong here.

6. A random variable has $p(1) = 0.2$, $p(2) = 0.5$, $p(5) = 0.3$. Find its mean and standard deviation. Is the mean a possible value, and does that matter?

7. You simulate a process 2 000 times and 350 of the runs end in failure. Estimate the failure probability and its simulation error, and say how many runs you would need to halve that error.

8. A colleague fits a normal distribution to the number of trucks arriving per hour, because the histogram looked bell shaped. Give two reasons to prefer a Poisson model, and one situation in which the normal would in fact be reasonable.

Answers to the odd-numbered exercises

1. $P(7) = 6/36 = 1/6$; $P(\geq 10) = 6/36 = 1/6$; $P(\text{at least one even}) = 1 - (3/6)(3/6) = 3/4$. The complement was easier for the third, where the direct count has 27 outcomes.

3. $P(0) = 0.96^{20} = 0.4420$; $P(1) = 20(0.04)(0.96)^{19} = 0.3683$; $P(2) = 190(0.04)^2(0.96)^{18} = 0.1458$, so $P(>2) = 1 - 0.4420 - 0.3683 - 0.1458 = 0.0439$. Mean $= 0.8$, $\sigma = \sqrt{20(0.04)(0.96)} = 0.876$.

5. $P(X > 75) = 0.0521$; $P(50 < X < 70) = 0.7745$; the top 10% exceed $62 + 1.2816(8) = 72.3$. The model is wrong because marks are bounded at 0 and 100, and usually skewed, so it assigns positive probability to impossible marks.

7. $\hat{p} = 350/2000 = 0.175$, with standard error $\sqrt{0.175(0.825)/2000} = 0.0085$, so roughly 0.175 ± 0.017. Halving the error needs four times the runs, that is 8 000.

Before next week

Simulate the birthday problem yourself for a class of your own size, with 10 000 runs, and compare your answer with the formula. Bring both numbers and your simulation error. Next week we condition: how probabilities change when you learn something.